Neil Patel

US Citizen | ndpatel@unc.edu | linkedin.com/in/neil-patel-735132281 | github.com/NeilP211

Education

University of North Carolina at Chapel Hill May 2028
B.S. in Computer Science *Chapel Hill, NC*
- Relevant Courses: Data Structures & Algorithms, Computer Systems, Software Design, Linear Algebra, Calculus 3

Experience

ASEA Brown Boveri (ABB) Cary, NC
Machine Learning Engineer Intern *Aug. 2026 – Present*
- Building the evidence case to switch on a dormant embedding-similarity routing tier in production: expanding the golden set with paraphrased, multilingual & adversarial queries to measure what it catches that rules miss
- Instrumenting per-agent latency, retry, failure-mode & token-cost telemetry across an 8-agent LLM pipeline

Software Engineering Intern *May 2026 – Aug. 2026*
- Engineered a real-time parts-replenishment platform on AWS replacing manual inventory-tracking across 5 production lines, recovering an estimated 900K parts/year ($9M in production value) by eliminating material-starvation
- Owned CI/CD pipelines deploying 2 interdependent services on every merge, removing manual release steps

SoberRide (Startup) Oct. 2024 – May 2025
Director of Engineering *Raleigh, NC*
- Co-architected a non-invasive facial intoxication classifier by fine-tuning an ImageNet-pretrained VGG-16 (Adamax) on 10K Haar-cascade-cropped face images, reaching 83% accuracy and beating a GPT-4o vision baseline
- Secured a U.S. patent; drove 400K+ views at CES 2025 (youngest startup), featured in IEEE Spectrum, won the Conrad/Diamond Challenge, endorsed by Mothers Against Drunk Driving, backed by Rep. Ross & Gov. Stein

NC State University Mar. 2023 – Mar. 2025
Lead Research Intern *Raleigh, NC*
- Published IEEE paper on skin cancer classification for minority demographics, increasing accuracy from 82% to 89%
- Roughly tripled the skin-tone-skewed dataset to Fitzpatrick IV–VI by masking lesions with a U-Net/R2U-Net+CBAM segmenter (0.79 IoU), engineering style transfer to background skin while preserving diagnostic detail

Projects

RLVR Evaluation Research | *Python, PyTorch, verl, AWS, SkyPilot*
- Showed the headline result of *Limit of RLVR* (Yue et al.) falls 3x (36 → 12 pts) under reworded test questions
- Measured an ensemble of 3 independently trained Qwen2.5 runs beating one model by 6% at matched sample budget
- Found MATH-500 is 98% saturated by the base models under test, leaving 10/500 items with usable signal

distkv | *Go, gRPC, Raft, bbolt, Kubernetes*
- Built a linearizable KV store on a from-scratch Raft engine: leader election, log replication, snapshots, joint consensus
- Served 9.9K linearizable reads/s at 5 ms P95 via ReadIndex vs. 82 writes/s on a 3-node cluster; 820 ms leader failover
- Verified 100% linearizability across 152K ops under 60s of seeded fault injection (5 chaos scenarios, WGL checker)
- Shipped 5 Prometheus metrics + Grafana dashboard, 3-replica K8s StatefulSet + Helm chart, GitHub Actions CI

Exfil | *Go, Rust/WASM, TypeScript, Python, PyTorch*
- Built a browser-playable, server-authoritative extraction shooter on a 64Hz Go sim with zero tick drops over a match
- Mirrored the sim in Rust (Q16.16 fixed-point); CI proves both sims produce identical world hashes over 2000 ticks
- Engineered client prediction & reconciliation (Rust/WASM) and lag-compensated hits validated at 50–150 ms ping
- Trained a self-play PPO bot to 100% win rate vs. random & scripted, exported to run natively in Go as the enemy AI

Publications

S. Chilaka, N. Patel, "Pass@k Is Not a Property of a Model," *Preprints*, 2026.
Kalra, Iyer, N. Patel, "Enhancing Skin Cancer Detection Across Skin Tones with Style Transfer Augmentation," *ACIT 2024*.
N. Patel, in G. Friedland (Ed.), *Information-Driven Machine Learning*, *Springer Nature*, 2024.

Technical Skills

Languages: Python, Go, Rust, TypeScript, JavaScript, Java, SQL Certifications: AWS Certified Cloud Practitioner
Libraries / Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face, FastAPI, axum, React, Next.js, WASM
Developer Tools: Docker, Kubernetes, Terraform, AWS, Azure, Linux, CI/CD, PostgreSQL, Snowflake, gRPC, Grafana
ML & AI: Triton, CLIP, RAG (hybrid retrieval, rerank), LLM agents (orchestration, HITL), LLM evals, RL (PPO)